



Interoperable Local-First in the age of AI

Hello, my name is Noel



- Developer & Entrepreneur.
- Working with the Solid Protocol.
- Working in the Open.

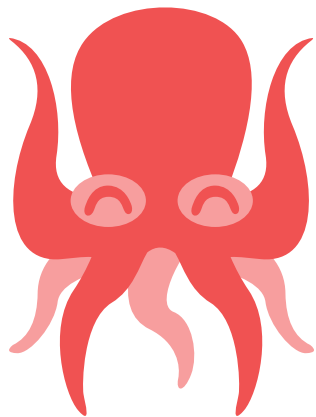
noeldemartin.com | noeldemartin.social | [@NoelDeMartin](https://twitter.com/NoelDeMartin)



My Apps



Focus



Media Kraken



Umai

Find them at noeldemartin.com/projects

My 3 Software Ideals

- Universality
 - The Web (any device)
 - Local-first (any location)

My 3 Software Ideals

- Universality
- Sovereignty
 - Longevity
 - Privacy & Data Ownership
 - Untangle Network Effects

My 3 Software Ideals

- Universality
- Sovereignty
- Interoperability
 - Malleable Software
 - Interoperable Serendipity

Interoperable Serendipity

“Interoperable Serendipity happens when apps built by different ~~developers~~ **Als** work together without explicit coordination”



noeldemartin.com/solid-world-2025

The Solid Protocol



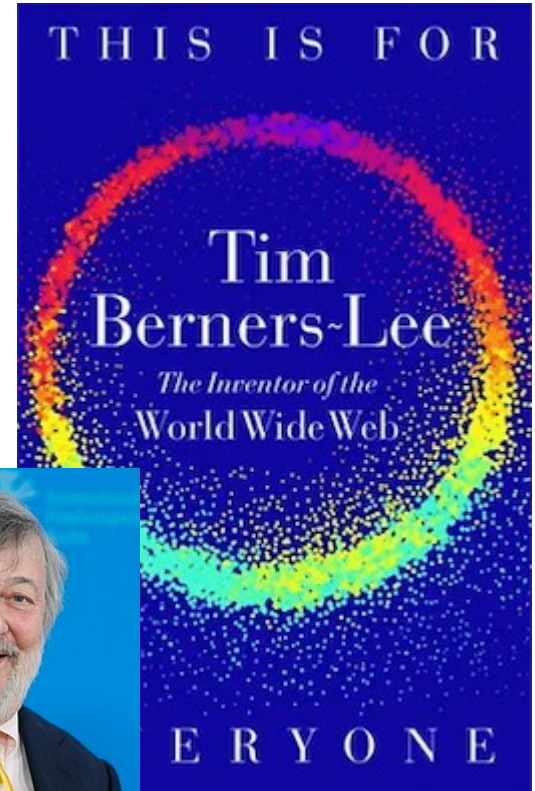
What is Solid?

- A protocol for decentralized storage.
 - “Decouple apps from data”

What is Solid?

- A protocol for decentralized storage.
- Created by Tim Berners-Lee
...as an evolution of the Web.

Narrated by



What is Solid?

- A protocol for decentralized storage.
- Created by Tim Berners-Lee.
- Learn more:
 - solidproject.org
 - [Why Solid?](#)

Solid Basic Concepts



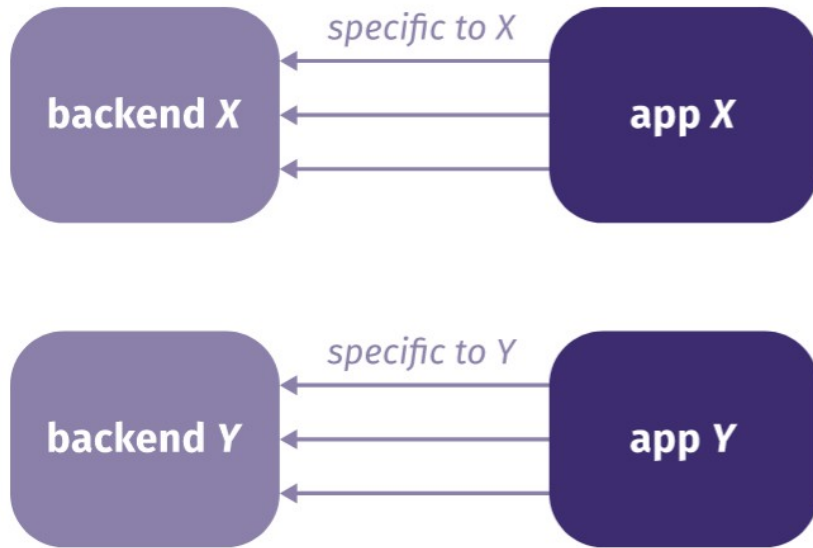
Solid Basic Concepts

- PODs → Personal Object Datastore

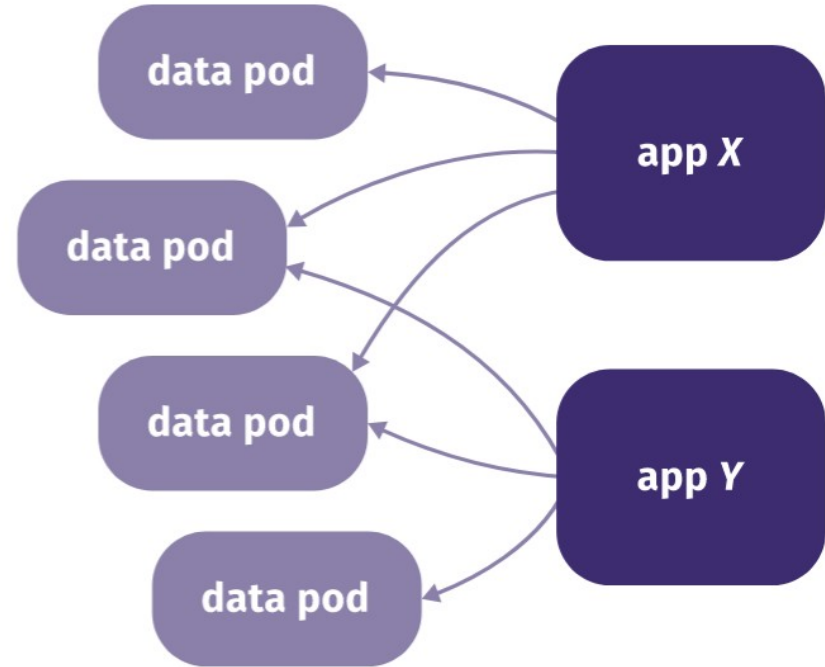


Solid Basic Concepts

centralized: single app & back-end



decentralized: multiple apps & back-ends



Solid Basic Concepts

- PODs → Personal Object Datastore
- RDF → Resource Definition Framework
 - Data Graph + Custom “Ontologies” or “Vocabularies”
 - e.g schema.org
 - Semantic Web

Solid Basic Concepts

- PODs → Personal Object Datastore
- RDF → Resource Definition Framework
- Built on top of the Web (GET, POST, etc.)
- Permissions, Authentication, etc.

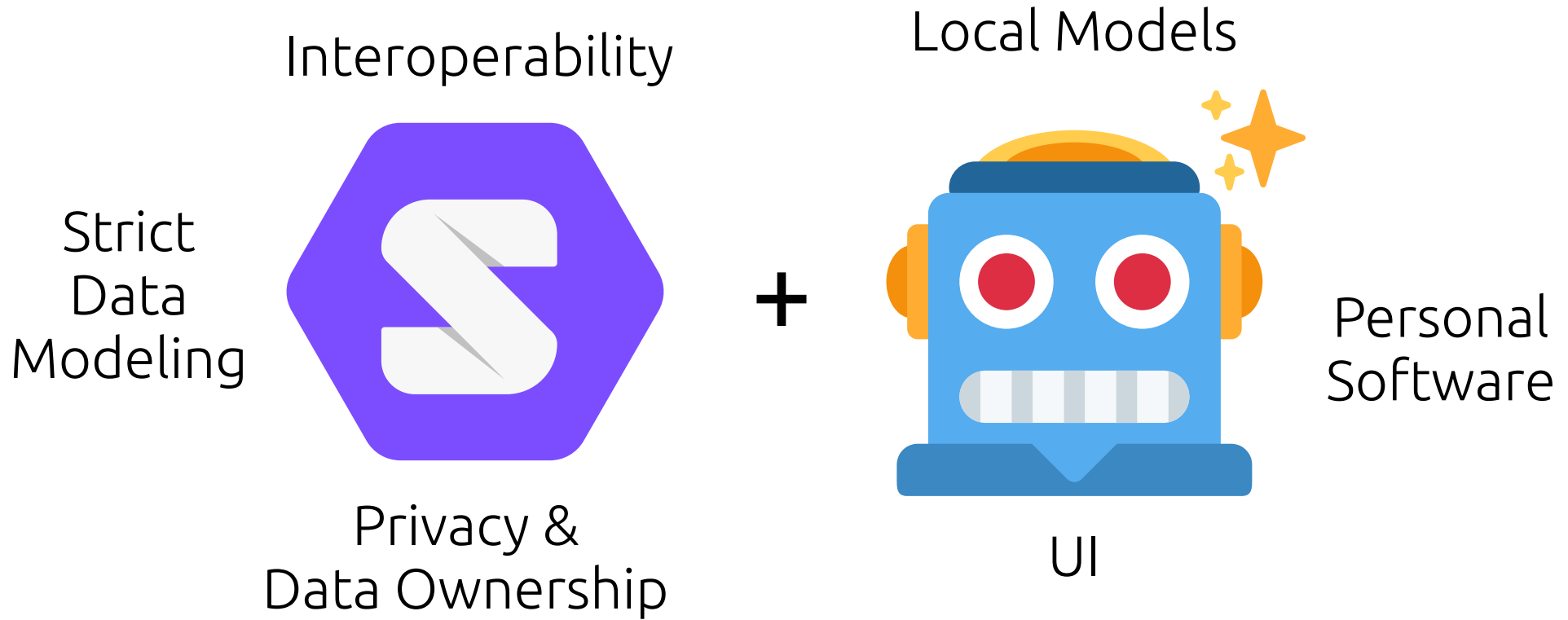


The Age of AI

- Generative AI
- Vibe Coding
- AI Agents
- ... but where are all the apps!?
- ... do we even want apps!?



Solid to the rescue?

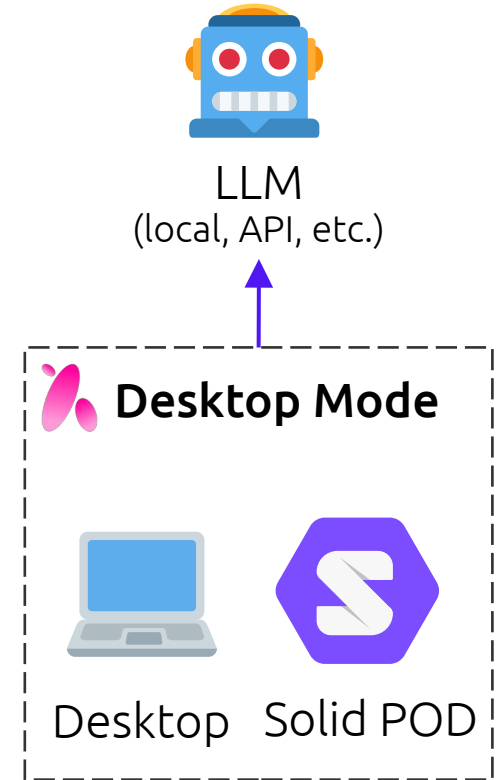
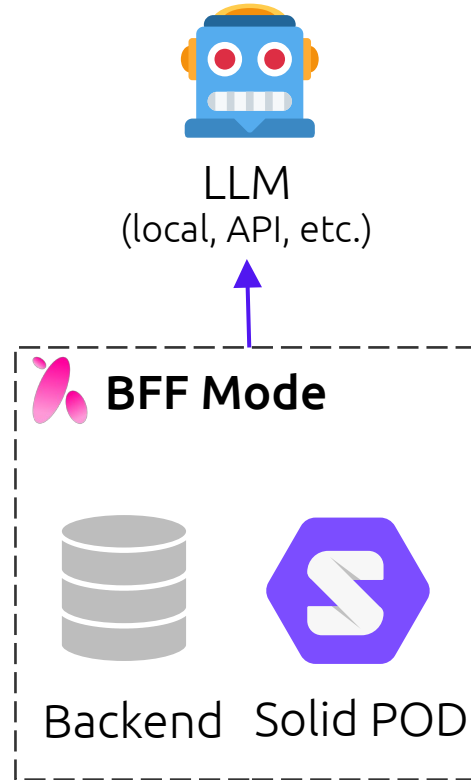
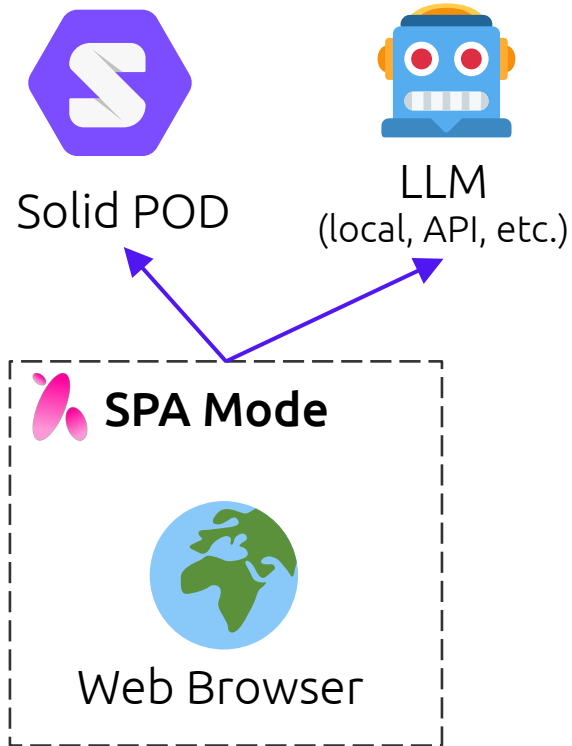




Inspired by “Charlie, the AI that works for you”

👉 See w3.org/DesignIssues/Charlie.html

Ànima Modes

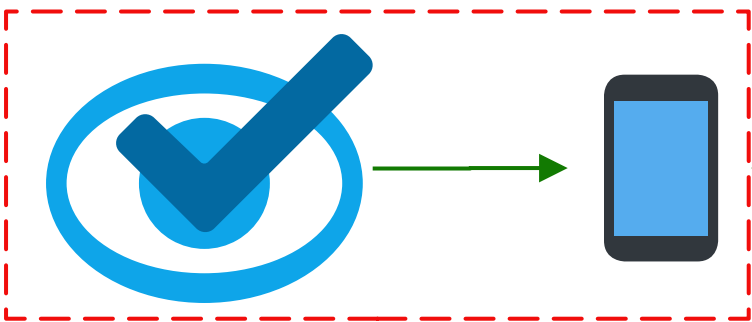


Ànima “Roadmap”

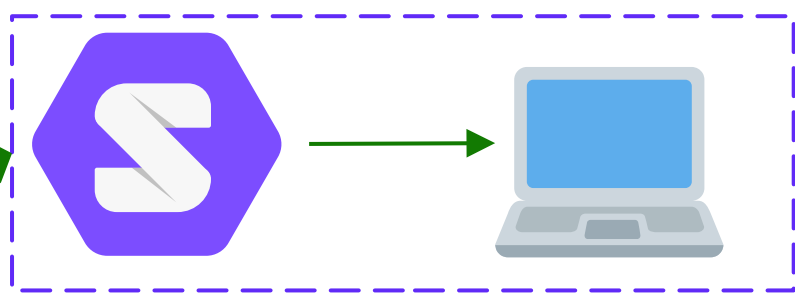
-  Model agnostic (Vercel’s AI SDK)
-  Read Solid POD
-  Built-in Solid POD (BFF/Desktop Modes)
-  External reachability (Desktop Mode)
-  Write to Solid POD
-  Create and serve Apps/Views
-  Agentic Loop
-  Scheduled and background Jobs
-  3rd-party chat integration (Telegram, WeChat, etc.)
-  Plugins / Tools

DEMO

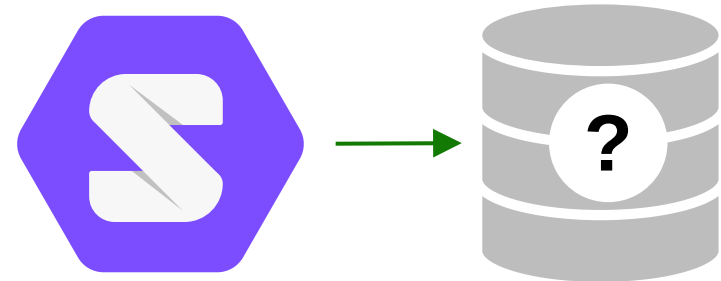
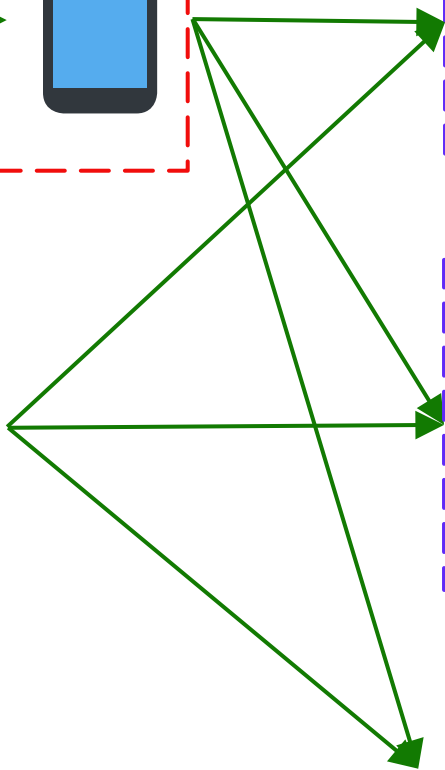
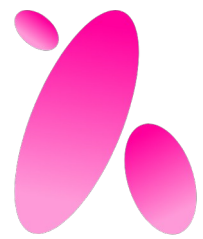
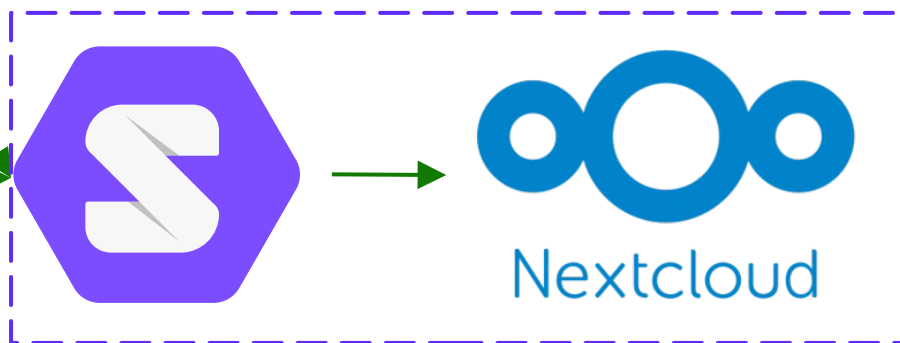
local-first



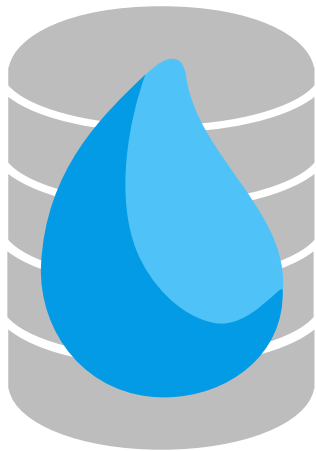
local network



self-hosted



My Tools



soukai.js.org

Framework agnostic

TypeScript

Models
(Active Record)

Engines
(Solid POD, IndexedDB, ...)

Interoperability
(Type Indexes, Schemas, RDF aliases)

Local-first
(CRDTs)

My Tools

Opinionated

(VueJS + TailwindCSS + Soukai)

UI Components

(Login, local-first, ...)

Authentication

Error handling

Cloud Sync

Aerogel

The Lightest Solid

Work in Progress!

aerogel.js.org

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My Tools

Thoughts on **Solid Developer Experience**

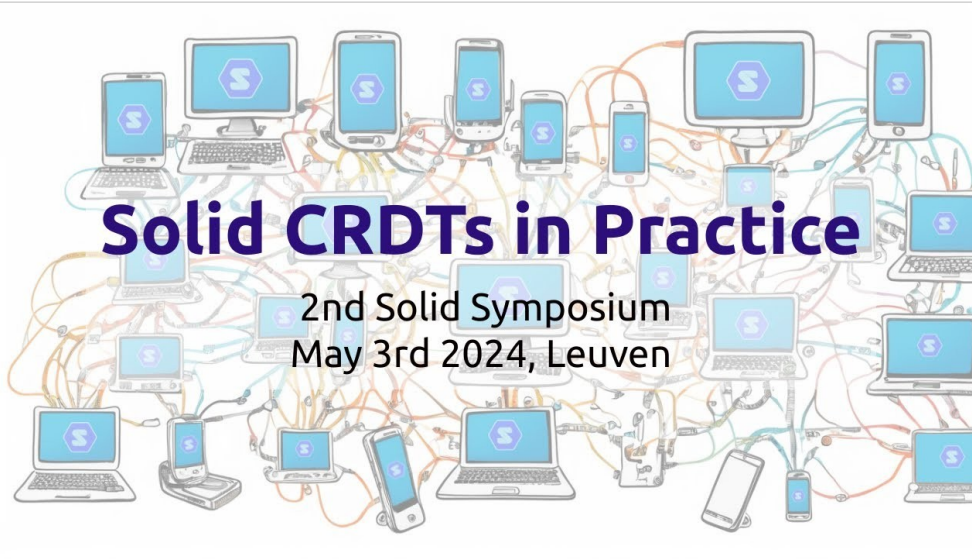


2nd Solid Symposium
May 2nd 2024, Leuven

noeldemartin.com/solid-symposium-dx

Solid CRDTs in Practice

2nd Solid Symposium
May 3rd 2024, Leuven



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